

Joe Cameron Blake Wolters

Pre-Medical Researcher — AI Medical Imaging, Clinical Data, and Digital Media

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EDUCATION

Bachelor of Science in Biology, Medical Concentration, University of South Florida, Tampa, FL — Expected May 2028

High School Diploma, Southwest Florida Christian Academy, Fort Myers, FL — 2018 to 2024

Graduated *Summa Cum Laude* · Weighted GPA: 4.02 · Unweighted GPA: 3.67

RESEARCH EXPERIENCE

Neurosurgery Research Assistant (Intern)

LSU Health Sciences Center Shreveport, Department of Neurosurgery — May 2025 to July 2025

- Contributed to an AI-driven MRI research project predicting Vertebral Bone Quality (VBQ) scores for preoperative spine surgery decision-making.
- Processed and organized over 1,000 MRI studies; manually reviewed and validated 400+ VBQ scores for publication preparation.
- Performed segmentation and image analysis using 3D Slicer, ITK-SNAP, and MicroDICOM in a structured clinical research workflow.
- Partnered with attending neurosurgeons, residents, and graduate researchers to maintain data accuracy and pipeline efficiency in a high-volume clinical research setting.
- Co-authored a manuscript currently under peer review at *Neurosurgery* (see Manuscripts Under Review section).

MANUSCRIPTS UNDER REVIEW / SUBMITTED

- Quinones CJ, Jameson W, Schwartz A, Lupo B, **Wolters J**, Kumbhare D, Guthikonda B, Hoang S. *Automated Quantification of Vertebral Bone Quality Score Using an Artificial Intelligence Framework*. *Neurosurgery*. Submitted September 2025.

TECHNICAL SKILLS

Medical Imaging & Research Tools: 3D Slicer (segmentation, MRI analysis), ITK-SNAP, MicroDICOM, manual VBQ scoring, clinical dataset validation.

Data & Productivity: Microsoft Excel (formulas, data cleaning, pivots), PowerPoint, Word; Google Workspace (Docs, Sheets, Drive); structured data entry and quality control on clinical datasets.

AI-Assisted Development & Tooling: Daily working knowledge of Anthropic Claude (including Claude Code as a development environment), OpenAI ChatGPT, and a custom Model Context Protocol (MCP) server (code -

review-graph) integrated into a personal production project. Comfortable using AI for research, prompting, "vibe coding," code review, and structured data tasks.

Web Development: HTML5, CSS3 (responsive design with mobile breakpoints), vanilla JavaScript, Firebase (Authentication, Firestore, Cloud Storage, Hosting), Git/GitHub, VS Code, Node.js / npm, Playwright (end-to-end browser testing), DNS and custom-domain configuration.

Creative & Media: CapCut (short- and long-form video editing), Adobe Photoshop, social media content creation and scheduling for YouTube and Instagram.

Domain Knowledge: Human anatomy, MRI fundamentals, imaging terminology, research data validation, clinical workflow exposure, semantic versioning, basic web security and deployment hygiene.

PROJECTS

Founder, Solo Developer & Maintainer — realanimereviews.com

Personal full-stack web project — 2024 to Present · Live at realanimereviews.com · Source: github.com/joewolters/real-anime-reviews

- Independently designed, built, and shipped a production web application from scratch: vanilla HTML, CSS, and JavaScript on the front end, Firebase (Auth, Firestore, Cloud Storage, Hosting) on the back end, deployed to a custom domain via Firebase Hosting and Namecheap DNS.
- Implemented user authentication (email/password, Google sign-in, password reset), per-user favorites and watchlists, an avatar upload pipeline (client-side image downscaling to 512px before Firebase Storage upload), real-time comments and voting, community reviews with discussion threads, official-rating votes, and a notifications inbox.
- Designed a five-collection-family Firestore data model with composite indexes powering a live "My Activity" feed; used onSnapshot listeners throughout so comments, votes, and notifications update without page refreshes.
- Built and maintain a 4,000-line main script (organized into 21 sectioned modules), a separate 780-line ES-module account page, responsive desktop/mobile stylesheets, and a hand-curated database of 50+ anime entries with cover art and metadata.
- Stood up an end-to-end test suite using Playwright (7 specs covering homepage load, search, modal lifecycle, memory-leak checks, deep-link routing, account-page load, and 404 handling) and codified a "tests must pass before any production-facing commit" rule for the project.
- Conducted a self-led code and security audit that surfaced and fixed real production issues, including a deploy-hygiene gap where files in `.gitignore` were still being uploaded by Firebase Hosting; codified a cross-system `".gitignore" ↔ "firebase.json ignore-array mirror"` rule to prevent recurrence.
- Maintain professional engineering documentation alongside the code: README, CHANGELOG (semantic versioning), ROADMAP, ARCHITECTURE, and DEPLOYMENT docs.
- Use AI development tooling intentionally and at depth: built the project largely with the help of Anthropic Claude Code and OpenAI ChatGPT, integrated a custom Model Context Protocol (MCP) server (code-review-graph) for graph-based code review and impact analysis, and maintain a project-level `CLAUDE.md` that codifies how AI agents should operate inside the repo.

Founder & Solo Developer — PickleClipper

Personal Node.js + FFmpeg + Python project — 2025 to Present · Source: github.com/joewolters/PickleClipper

- Solo-building a local web app that auto-trims pickleball footage to just the rally action, replacing a manual editing step that was eating hours per match.
- Designed and shipped a custom spectral-flux audio-detection algorithm with solo-peak rally clustering; achieves ~92% recall across labeled outdoor singles clips (F1 0.768–0.881 on the held-out test set).
- Currently integrating Ultralytics YOLOv8 for player tracking as a visual confirmation layer on top of the audio detector.
- Wrote a parameterized test harness (`npm run test-detection`) that runs against a labeled ground-truth dataset and emits per-clip F1, precision, and recall metrics, plus an autotune harness for sweeping algorithm settings.

Founder, Video Editor & Content Creator — lowlvledits

YouTube and Instagram (@lowlvledits) — Active

- Produces and edits short-form and long-form video content for YouTube and Instagram under the @lowlvledits brand.
- Owns the full production pipeline: footage selection, editing in CapCut, color and audio adjustments, captioning, thumbnails, posting, and audience engagement.
- Hands-on experience with the kind of multi-clip, multi-track editing workflows used in 2D & 3D digital media production.

CLINICAL EXPERIENCE

Medical Shadowing — Emergency Medicine

Archbold Hospital, Thomasville, GA

- Shadowed emergency room physicians during patient assessments, critical decision-making, and emergency procedures.
- Observed triage, patient stabilization, and multidisciplinary collaboration in a high-acuity hospital environment.
- Developed familiarity with physician–patient communication, clinical professionalism, and ER workflow.

LEADERSHIP & ACTIVITIES

Risk Manager, Executive Board — USF Running Club

University of South Florida, Tampa, FL — Spring 2025 to Present

- Elected to the Executive Board to oversee safety, risk management, and emergency response for all club activities.
- Maintain active CPR certification per the USF Running Club Constitution.
- Attend practices and traveling events to ensure adherence to lightning policy, serious-incident protocol, and traffic-safety standards.
- Complete and submit accident reports; ensure full membership compliance with USF Sports Club participation waivers.
- Coordinate with USF administration on event compliance, emergency planning, and travel logistics.

- Lead fundraising for the club's 501(c)(3) account, including partner-restaurant outreach, fundraiser applications, and event coordination.
- Collaborate with the rest of the E-Board to plan races, practices, and travel for a multi-semester sports club.

Captain — Cross-Country and Track & Field

Southwest Florida Christian Academy, Fort Myers, FL — 2020 to 2024 (4 consecutive years)

- Served as team captain for both Cross-Country and Track & Field across all four years of high school competition.
- Led practices, set the tone for team culture, and mentored underclassmen on training, racing strategy, and competition mindset.
- Coordinated logistics with coaches and represented the team at meets and championships throughout Southwest Florida.

WORK EXPERIENCE

Event Server — Good Food Catering Company

Tampa, FL — September 2024 to Present

- Provide professional food and beverage service at private and corporate events of varying scale.
- Coordinate event setup, service, and breakdown to ensure smooth, high-quality guest experiences.
- Collaborate with team members to manage large-scale functions under tight timelines, maintaining professionalism, precision, and client satisfaction in fast-paced environments.

Warehouse Supervisor — Sensible Spa Solutions

Fort Myers, FL — January 2019 to 2023

- Managed inventory organization and product restocking for spa equipment, ensuring accurate orders and timely fulfillment.
- Filed and processed warranty service claims, prepared product returns, and coordinated with vendors and customers.
- Supported general office operations during peak periods.

VOLUNTEER & COMMUNITY SERVICE

- **USF Bull Haul Move-In Volunteer**, University of South Florida, 2024 to 2025 — assisted incoming students and fa